

AERO 40009 Thermodynamics and Heat Transfer
Tutorial Sheet 5 (Heat Transfer)

Question 1.

Calculate the average temperature of the surface of Mercury, assuming that it behaves as a black body of radius 2440 km, at a distance of 5.79×10^7 km from the sun.

Take the radius of the sun to be 6.96×10^5 km, with a surface temperature of 5760K and treat it also as a black body. Assume that the surface of mercury loses heat all over by re-radiation to deep space, which is effectively at zero degrees. Look up the correct value and compare with your result.

Hints – similar to calculation for earth:

- What is the amount of solar energy which could be absorbed on a square metre of surface at a distance corresponding to the mean radius of the orbit of the mercury?
- What is the projected area of the mercury (which receives the solar radiation)?
- T surface must balance absorbed/radiated.

$$\dot{E}_{\text{sun}} = A_{\text{sun}} \sigma T_{\text{sun}}^4 = (4\pi r_{\text{sun}}^2) \sigma T_{\text{sun}}^4 = 3.79928 \times 10^{26} \text{ J/s}$$

$$\dot{q}_{\text{avg}} = \frac{\dot{E}_{\text{sun}}}{4\pi r_{\text{orbit}}^2} = 9018.5 \text{ J/m}^2 \cdot \text{s}$$

$$\therefore \dot{E}_{\text{received}} = \dot{q}_{\text{avg}} \cdot A_{\text{mer}} = \dot{q}_{\text{avg}} \cdot (\pi r_{\text{mer}}^2) = 1.6868 \times 10^7 \text{ J/s}$$

$$\therefore \dot{E}_{\text{received}} - \dot{E}_{\text{rad}} = 0$$

$$\therefore \dot{E}_{\text{received}} - \epsilon \sigma A (T_{\text{surface}}^4 - T_{\text{space}}^4) = 0$$

$$\therefore T_{\text{surface}} = \sqrt[4]{\frac{\dot{E}_{\text{received}}}{\epsilon \sigma A_{\text{mercury}}}} = 446.55 \text{ K}$$

Question 2

Water flows through a long thin walled tube, of internal diameter 2 cm. The outer walls of the tube are heated by means of an electric current through a wire, which is uniformly and tightly wrapped around the tube.

(a) If at a certain point far from the inlet, the bulk water temperature is 75 °C, and the wall temperature is 95 °C, what is the heat transfer rate to the water, given that the flow rate through the tube is 5 litres/min? (Answer: 44.6 kW/m²)

Hints: (a)

- Is the flow likely to be laminar or turbulent? Compute Reynolds no.
- What is the appropriate expression to find the Nusselt no. from data sheet?
- Calculate Nu → hence find heat transfer coefficient, h; $q = h\Delta T$

$$\begin{aligned} \text{(a)} \quad Re &= \frac{\rho u D}{\mu} = \frac{973.15 \text{ kg/m}^3 \times \left(\frac{5 \text{ L/min}}{\pi (1 \text{ cm})^2} \right) \times (2 \text{ cm})}{4.1465 \times 10^{-4} \text{ Pa.s}} \\ &= 12450.79 > 10000 \end{aligned}$$

∴ flow is more likely to be turbulent

$$\therefore Nu = \frac{hL}{k} \approx 0.022 Pr^{\frac{1}{2}} Re^{0.8} = 0.022 \left(\frac{C_p \mu}{k} \right)^{0.5} Re^{0.8}$$

$$\begin{aligned} \therefore h &= \frac{k}{L} \cdot 0.022 \left(\frac{C_p \mu}{k} \right)^{0.5} Re^{0.8} & * k &= 0.6585 \\ &= 2224 \text{ W/Km}^2 & C_p &= 4.1985 \times 10^3 \end{aligned}$$

$$\therefore \dot{q} = h\Delta T = 44.488 \text{ kW/m}^2$$

(b) If the flow rate were reduced to 0.25 litres/min, and the current flow adjusted, so that the bulk temperature of the water, and the wall temperature were unchanged at this same point, how would the heat transfer rate be affected? Explain

$$\therefore \text{b) } Re' = Re \frac{0.25}{5} = 622.5395$$

$\therefore$ The flow becomes turbulent

$$\therefore Nu = \frac{hL}{k} = 3.66$$

$$\therefore h = \frac{Nuk}{L} = 120.5055 \text{ W/m}^2\text{K}$$

$$\therefore \dot{q} = h\Delta T = 2.41011 \text{ kW/m}^2$$

Question 3

Taking the Stefan Boltzmann constant as $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$, find the effective heat transfer coefficient between a surface at 300 K and surroundings at 280 K. Treat all surfaces as black bodies. (Ans. $5.53 \text{ W/m}^2\text{K}$)

$$h \Delta T = \epsilon \sigma (T_{\text{surface}}^4 - T_{\text{surroundings}}^4)$$

$$\therefore h = \frac{\epsilon \sigma (T_{\text{surface}}^4 - T_{\text{surroundings}}^4)}{\Delta T}$$

$$= 5.538 \text{ W/m}^2\text{K}$$

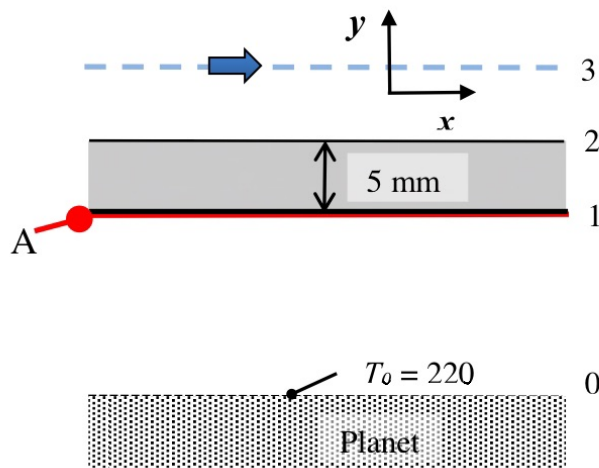
Question 4. This is a long question covering nearly every aspect of heat transfer!

Figure 1 shows an idealised model of a device mounted on a satellite in a low orbit about a planet. The device extends a length in the horizontal (x) direction many times its thickness. The device comprises:

- an electronic component “A” of negligible thickness, (at position 1 in the figure), dissipating heat at a rate $q_d = 10 \text{ W.cm}^{-2}$, and bonded to a
- metal plate, of thickness $\delta = 5 \text{ mm}$, which is cooled by
- a flow in a channel, (region above level 2 in the figure). The exposed surface of A exchanges heat by radiation with the planetary surface, which is at a temperature $T_0 = 220 \text{ K}$.

Metal plate properties: $k = 300 \text{ W.m}^{-1}\text{K}^{-1}$; $\rho = 3.0 \times 10^3 \text{ kg.m}^{-3}$; $C = 1000 \text{ J.kg}^{-1}\text{K}^{-1}$

Coolant flow convective heat transfer coefficient: $h_c = 1000 \text{ W.m}^{-2}\text{K}^{-1}$



- (a) Identify the elements contributing to thermal resistance in the normal (y) direction and write out an expression for the heat transfer across each element. Treat the exposed surface of A and of the planet as black bodies.

$$1 : \dot{q}_T = 10 \text{ W/cm}^2 = 1 \times 10^5 \text{ W/m}^2$$

$$1 \rightarrow 2 : \dot{q}_m = k \frac{dT}{dx} = 300 \text{ W/mK} \frac{\Delta T}{5 \times 10^{-3} \text{ m}} = 6 \times 10^4 (T_1 - T_2)$$

$$2 \rightarrow 3 : \dot{q}_f = h_c \Delta T = 1000 \Delta T = 1000 (T_2 - T_3)$$

$$0 \rightarrow 1 : \dot{q}_p = \epsilon \sigma (T_0^4 - T_1^4) = \sigma (T_0^4 - T_1^4)$$

$$\dot{q}_m = \dot{q}_f = k \frac{T_1 - T_2}{\delta} = h_c (T_2 - T_3) = \dot{q}_c$$

- (b) Write out an expression relating total power dissipated and the temperatures of device, T_1 , of bulk coolant temperature, T_3 and of the planet, T_0 .

$$\dot{q}_T = \dot{q}_m + \dot{q}_f + \dot{q}_p$$

$$= k \frac{T_1 - T_2}{\delta} + h_c (T_2 - T_3) + \epsilon (T_0^* - T_1^*)$$

$$\therefore T_1 - T_3 = (T_1 - T_2) + (T_2 - T_3)$$

$$= \dot{q}_c \left(\frac{\delta}{k} \right) + \dot{q}_c \left(\frac{1}{h_c} \right) = \dot{q}_c \left(\frac{\delta}{k} + \frac{1}{h_c} \right)$$

$$\therefore \dot{q}_T = \dot{q}_c + \dot{q}_p$$

$$= \frac{T_1 - T_3}{\frac{\delta}{k} + \frac{1}{h_c}} + \epsilon (T_1^* - T_0^*)$$

- (c) If T_1 is 260 K at a certain distance x_l along A, estimate the bulk coolant temperature at the corresponding location.

$$T_o = 220K$$

$$\therefore \dot{q}_p = 126.28224W$$

$$\therefore \dot{q}_c = \dot{q}_T - \dot{q}_p = 99873.71776 W/m^2$$

$$\therefore \frac{260K - T_3}{\frac{5 \times 10^{-3}m}{300 W/mK} + \frac{1}{1000 W/mK}} = 99873.71776 W/m^2$$

$$\therefore T_3 = 158.46K$$

- (d) Neglecting heat transfer by radiation, show that the increase in coolant temperature over a length L in the x -direction is given by:

$$\Delta T = \frac{(T_1 - T_3)L}{\dot{m}_w C \left[\frac{1}{h} + \frac{\delta}{k} \right]}$$

where $\dot{m}_w$ is the coolant mass flow rate per unit width of the channel.

Neglecting radiation: $\dot{q}_r = \dot{q}_c$

$$\begin{aligned} \therefore \dot{q}_r &= \frac{T_1 - T_3}{\left(\frac{1}{h} + \frac{\delta}{k} \right)} = \frac{\dot{m} C \Delta T}{A} = \frac{\dot{m} C \Delta T}{dz \cdot L} \\ &= \frac{\dot{m}_w \cdot dz \cdot C \Delta T}{dz \cdot L} = \frac{\dot{m}_w C \Delta T}{L} \end{aligned}$$

$$\therefore \Delta T = \frac{(T_1 - T_3)L}{\left(\frac{1}{h} + \frac{\delta}{k} \right) \dot{m}_w C}$$

- (e) In the event of a sudden loss of coolant, the channel is assumed to act as an insulator, so that heat loss from the device is confined to that due to radiation from A.

Estimate the initial rate of temperature rise of A, and the time taken for the device to reach 400 K, treating it together with the attached metal plate as a single lumped element.

$$mC \frac{dT}{dt} = \dot{q}_T A - \dot{q}_P A = \rho V C \frac{dT}{dt} = \rho A \delta C \frac{dT}{dt}$$

$$\therefore \frac{dT}{dt} = \frac{\dot{q}_T - \dot{q}_P}{\rho \delta C} = 6.658 \text{ K/s}$$

$$\therefore t = \frac{\Delta T}{dT/dt} = 21.03 \text{ s}$$

Question 5

A hot body is suddenly immersed in a cold fluid, which is at temperature T_0 . After some time, the gradient of temperature across the body is assumed to be of order:

$$(T_i - T_0)/r_0 ,$$

where T_i is the temperature at its core, T_0 is the temperature its surface, and r_0 represents an effective radius of the body.

- (i) Write down the equation governing heat transfer by conduction across the body, and the corresponding equation governing convection at the surface.

$$\dot{q}_c = k \frac{dT}{dx} = k \frac{T_i - T_0}{r_0} \quad (\text{when } T_s = T_0 \text{ after some time})$$

$$\dot{q}_s = h \Delta T = h (T_0 - T_\infty)$$

- (ii) By equating the convective and conductive heat transfers, show that

$$T_i - T_o \approx \frac{Bi}{1 + Bi} (T_i - T_\infty),$$

where $Bi = (hr_o/k)$ is the Biot number, and h, k represent the conductivity and convective heat transfer coefficient respectively.

When $\dot{q}_c = \dot{q}_s$

$$k \frac{T_i - T_o}{r_o} = h (T_o - T_\infty)$$

$$\therefore T_i - T_o = Bi (T_o - T_\infty)$$

$$\therefore T_i - T_\infty = (T_i - T_o) + (T_o - T_\infty)$$

$$= (T_i - T_o) + \frac{1}{Bi} (T_i - T_o)$$

$$= \frac{Bi + 1}{Bi} (T_i - T_o)$$

$$\therefore T_i - T_o = \frac{Bi}{Bi + 1} (T_i - T_\infty)$$

(iii) What is the significance of a low value of the Biot number?

Nearly uniform temperature

Question 6

Calculate the U value for a cavity wall consisting of:

- 102 mm brick external wall (thermal conductivity $k = 0.77 \text{ W/mK}$)
- 50 mm air cavity
- 100 mm block ($k = 0.57 \text{ W/mK}$)
- 13 mm plaster on inner surface ($k = 0.18 \text{ W/mK}$)

The standard value of thermal resistance for a clear, non-mirrored internal cavity can be taken as $0.18 \text{ m}^2\text{K/W}$; the thermal resistance on the external (exposed) and internal surfaces is taken as $R_{Se} = 0.04 \text{ m}^2\text{K/W}$ and $R_{Si} = 0.13 \text{ m}^2\text{K/W}$ respectively.

(Ans. $1.37 \text{ m}^2\text{K/W}$)

(For those wanting to know more about construction or planning their own building, see Ref. CIBSE guide A, BS ISO 10456:2007 Building materials and products)

$$\begin{aligned}\frac{1}{U_T} &= \frac{1}{U_{Ex}} + \frac{1}{U_{Gr}} + \frac{1}{U_A} + \frac{1}{U_{Bl}} + \frac{1}{U_P} + \frac{1}{U_{In}} \\ &= R_{Ex} + \frac{x_{Br}}{k_{Br}} + R_A + \frac{x_{Bl}}{k_{Bl}} + \frac{x_P}{k_P} + R_{In} \\ &= 0.730 \text{ W/m}^2\text{K}\end{aligned}$$

$$\therefore U_T = 1.370 \text{ m}^2\text{K/W}$$

Question 7

(a) Show that the heat transfer rate by conduction through a thick walled tube, of inner radius R_i and outer radius R_o is related to the temperature difference ΔT across the tube wall according to:

$$\dot{q}_i = U(T_i - T_o) \quad ; \quad U = \frac{k}{R_i \ln(R_o/R_i)}$$

$$\rho C \frac{\partial T}{\partial t} = k \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) \right] + S_v$$

(b) Water at high pressure flows rapidly through a tube of inner diameter 2 cm, outer diameter 3 cm. At some point along the tube, the temperature of the inner wall is 80 °C and that of the outer wall is 50 °C. Take the thermal conductivity of the tube material to be $400 \text{ Wm}^{-1}\text{K}^{-1}$. Calculate the heat transfer rate per unit area.

(c) If the water flows at 10 kg/s and the tube is 5 metres long, estimate the temperature rise/fall of the water. (Take the specific heat of water to be 4.2 kJ/kg). Assume constant heat flux, and that the water enters at a temperature of 80 °C. Comment on the result.